

Project5

需要对比标准答案 Translation

```
star@star-A:~/Desktop/RoboticBasic/Lecture5$ ros2 run tf2_ros tf2_echo rx150/base link rx150/wrist link
At time 1784886295.14349034
- Translation: [-0.091, -0.052, 0.373]
- Rotation: in Quaternion (xyzw) [0.183, -0.683, 0.183, 0.683]
- Rotation: in RPY (radian) [0.464, -1.571, 0.000]
- Rotation: in RPY (degree) [26.565, -90.000, 0.000]
- Matrix:
  0.000 -0.500 -0.866 -0.091
  0.000  0.866 -0.500 -0.052
  1.000  0.000  0.000  0.373
  0.000  0.000  0.000  1.000
At time 1784886296.14346120
- Translation: [-0.091, -0.052, 0.373]
- Rotation: in Quaternion (xyzw) [0.183, -0.683, 0.183, 0.683]
- Rotation: in RPY (radian) [0.464, -1.571, 0.000]
- Rotation: in RPY (degree) [26.565, -90.000, 0.000]
- Matrix:
  0.000 -0.500 -0.866 -0.091
  0.000  0.866 -0.500 -0.052
  1.000  0.000  0.000  0.373
```

以下是 3 种方法对比

```
star@star-A:~/Desktop/RoboticBasic/Lecture5$ python3 test_forward_kinematics.py
[INFO] [1784886628.422115068] [manipulator]: Press Ctrl + C to terminate
[INFO] [1784886628.423758156] [manipulator]: Joint Angle Test Case: [0.5235987755982988, -1.0471975511965
976, -0.5235987755982988]
[INFO] [1784886628.423888246] [manipulator]: Method          x          y          z
[INFO] [1784886628.423998984] [manipulator]: -----
[INFO] [1784886628.424116314] [manipulator]: Trigonometry    -0.0908    -0.0525    0.3723
[INFO] [1784886628.424229935] [manipulator]: D-H Parameters  -0.0908    -0.0525    0.3723
[INFO] [1784886628.424341332] [manipulator]: PoE (Exponential) -0.0908    -0.0525    0.3723
```

X,Y,Z 和 Translation 算出来是一样的